INCH-POUND

MIL-DTL-23554E(SH)

5 November 2009

SUPERSEDING

MIL-W-23554D(SH)

31 March 1994

DETAIL SPECIFICATION

WASHER-EXTRACTOR, LAUNDRY, 16- TO 20-POUND CAPACITY (SUBMARINE SERVICE)

Reinstated after 5 November 2009 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 16- to 20-pound dry weight capacity automatic combination washer-extractor for use aboard submarines.

1.2 Classification. Washer-extractors are of the following types, as specified (see 6.2).

Type I – Shock qualified (see 3.5.5 and 6.1)

Type II – Not shock qualified (see 3.5.5 and 6.1)

1.3 Part or Identifying Number (PIN). PINs to be used for washer-extractor acquired to this specification are created as follows:

M	23554	-	X
Prefix for Military Specification	Specification Number		Type (see code below)

Code		Type
1	-	Type I: Shock qualified
2	-	Type II: Not shock qualified

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATION

MIL-S-901	-	Shock Tests. H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-740-1	-	Airborne Sound Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements Acceptance Criteria Of Shipboard Equipment
MIL-STD-1399-300	-	Electric Power, Alternating Current

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME INTERNATIONAL

ASME B1.20.1	-	Pipe Threads, General Purpose (Inch). (DoD adopted)
ASME B1.20.7	-	Hose Coupling Screw Threads – Inch. (DoD adopted)

Copies of this/these document/s are available from ASME International, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.

ASTM INTERNATIONAL

ASTM B43	-	Pipe, Standard Sizes, Seamless Red Brass. (DoD adopted)
ASTM B61	-	Castings, Steam or Valve Bronze. (DoD adopted)
ASTM B62	-	Metal Castings, Composition Bronze or Ounce. (DoD adopted)
ASTM B75	-	Tube, Seamless Copper. (DoD adopted)
ASTM A123/A123M	-	Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products. (DoD adopted)
ASTM A167	-	Steel Plate, Sheet, and Strip, Stainless and Heat-Resisting Chromium-Nickel. (DoD adopted)
ASTM A269	-	Tubing for General Service, Seamless and Welded Austenitic Stainless Steel. (DoD adopted)
ASTM D3359	-	Standard Test Methods for Measuring Adhesion by Tape Test.
ASTM D3363	-	Standard Test Method for Film Hardness by Pencil Test. (DoD adopted)

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 45 - Electric Installation on Shipboard. (DoD adopted)

(Copies of these documents are available from the Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331 or online at www.ieee.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials.

3.3.1 Structure. Frame, base, and strength members shall be constructed of malleable iron, ductile iron, or steel, and shall be hot-dipped galvanized or primed and painted. Gray cast iron or other brittle materials shall not be used for these parts.

3.3.2 Shell. Unless otherwise specified, shell shall be made of corrosion-resistant steel in accordance with ASTM A167, type 302 or 304. Front of the shell shall be corrosion-resistant steel or ferrous metal with corrosion-resisting metal liners on the side exposed to the washing solution. The shell shall be provided with a corrosion-resistant steel door frame.

3.3.3 Cylinder. The cylinder shall be made of corrosion-resistant steel in accordance with ASTM A167, type 302 or 304. The driving trunnion and spider shall be the same material as the cylinder, or ferrous metal protected by either hot-dip galvanizing or a corrosion-resistant coating.

3.3.4 Cabinet. The cabinet shall be made of corrosion-resistant steel in accordance with ASTM A167, type 302 or 304.

3.3.5 Metal joints. Rivets, bolts, screws, nuts, and washers shall be of steel, except when brass or corrosion-resistant steel is fastened, in which case they shall be of brass or corrosion-resistant steel. Where dissimilar metals are fastened, rivets, bolts, screws, nuts, and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal of the parts being welded. Welded steel parts shall be galvanized after completion of welding in accordance with ASTM A123/A123M. Nuts subject to vibration or shock shall be self-locking. Drill point screws are prohibited. Spacing of fasteners shall provide strength, tightness, and rattle-free performance. Convenient access shall be provided to fasteners of parts requiring service.

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3.3.6 Pipes, fittings, and valves. Parts in contact with the wash solution shall be bronze or corrosion-resistant steel. The following materials shall be used:

Pipes:	Corrosion-resistant steel, in accordance with ASTM A269, Grade TP 302 or 304.
	Brass, in accordance with ASTM B43, Grade A.
	Copper, in accordance with ASTM B75.
Fittings:	Bronze, in accordance with ASTM B61 or ASTM B62.
Valves:	Cast bronze, in accordance with ASTM B61.
	Plastic, in accordance with the manufacturer's commercial practice.

3.3.7 Drain assembly. The drain assembly shall be made of the following materials:

Drain valve body:	Cast bronze, in accordance with ASTM B61.
	Plastic, in accordance with the manufacturer's commercial practice.
Drain fittings:	Cast bronze, in accordance with ASTM B61 or ASTM B62.
	Plastic, in accordance with manufacturer's commercial practice.
Drain piping:	Corrosion-resistant steel, in accordance with ASTM A269, Grade TP 302 or 304.
	Brass, in accordance with ASTM B43 Grade A.
	Copper, in accordance with ASTM B75.
	Rubber hose, in accordance with the manufacturer's commercial practice.

3.3.8 Restricted materials. The washer-extractor shall be free from asbestos, wood, and cadmium materials.

3.4 Configuration. The washer-extractor shall be front loading, automatic, shell and cylinder (open pocket) type with a rotating and reversing action about its horizontal axis. Each machine shall consist of a remote control box, a washer-extractor enclosed within a cabinet, and the necessary controls, valves, motor, and appurtenances for automatically washing, rinsing, and extracting clothing as specified herein. The remote control box shall consist of the automatic program control, and all other components and external cabling required to control washer-extractor operation from a remote location.

3.4.1 Special design requirements. The washer-extractor shall be so constructed that it can be readily disassembled into sections or parts that can be passed through a 30-inch diameter by 60-inch long cylinder and reassembled with a minimum of effort in the submarine laundry space. Cutting of parts by torch or similar means to accomplish disassembly at the site, and reassembly of parts by welding, riveting, or similar method shall not be permitted. Internal wiring to components requiring disconnection during disassembly shall use plug type connectors to allow for easy removal and installation of component parts without the use of special tools. Not less than 50 percent of the original hardware and sound dampening material shall be provided with each machine as spare material to aid reassembly.

3.4.1.1 Capacity. The washer-extractor dry weight load capacity shall be not less than 16 pounds and shall be not greater than 20 pounds preload cotton clothing (see 6.2).

3.4.2 Size. Excluding appendages such as the door handle, knobs, etc, the washer-extractor overall dimensions shall be not greater than 24½ inches wide by 26 inches deep by 34 inches high.

3.4.3 Weight. The washer-extractor weight, including controls, remote control box, and appurtenances, shall be not greater than 460 pounds.

3.4.4 Water consumption. Water consumption shall be not greater than 50 gallons for the maximum dry weight load capacity.

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3.4.5 Mounting. The washer-extractor shall be mounted to a deck foundation. The remote control box shall be capable of being both shelf mounted and bulkhead mounted (see 3.4.14.1). Type I and Type II washer-extractors shall be provided with a minimum of four bolt holes located on the machine base for securing the unit to a deck foundation. Bolt holes shall be accessible from the front and rear of the machine. The Type I washer-extractor and remote control box bolt holes and bolts shall keep the washer-extractor secured to the deck and the remote control box secured to the shelf or bulkhead when subjected to the shock requirements specified in 3.5.5. Type II washer-extractor bolt holes shall be sized for a ⅝-inch bolt.

3.4.6 Shell. Except for required shell openings, the shell shall be watertight. When closed or in place, the door, drain valve, water inlet valves, cylinder shaft, water level sensor, and overflow connection shall be installed to provide a watertight connection to the shell. The shell shall be rigid to prevent distortion, rattling, or objectionable vibration under normal shipboard operation. Reinforcement shall be provided where necessary to properly support attachments. Shell seams shall not pass through the drain valve opening.

3.4.6.1 Shell openings. The shell shall have openings for a door, drain valve, water inlet valves, cylinder shaft, water level sensor, and overflow connection. No other openings shall be provided.

3.4.7 Cylinder. The cylinder shall be rigid to prevent distortion and failure from stresses caused during the extraction cycle at the maximum dry weight load capacity. Perforations in the cylinder shall be sized and spaced for efficient washing, rinsing, and extraction. Perforations shall be free from burrs or imperfections that can damage clothing. The washer-extractor shall be constructed so the cylinder can be replaced.

3.4.8 Cabinet. The top of the washer-extractor cabinet shall be round. Front and rear panels shall be completely removable by means of mechanical fasteners in order to access the front and back of the unit for installation, maintenance, and repair. Metal-to-metal contact between framing, sheet metal, and other items that may cause rattling or a noise short shall be isolated by the use of sound dampening material.

3.4.9 Door. The shell shall be fitted with a round door mounted on heavy duty hinges, complete with neoprene gasket and heavy duty latch to ensure watertightness when the door is closed. When the door is fitted with a window, the window shall be made of tempered glass of a quality and thickness to prevent damage under normal operation. The glass shall be sealed and capable of ready replacement.

3.4.10 Drain valve. The drain valve shall have a resilient seat and shall be fitted at the lowest point of the shell. The valve shall be normally open. The drain valve shall be 2 inches IPS (Iron Pipe Size) and shall be located on the lower back of the machine not more than 6 inches off of the center of the machine. The center height of the drain outlet location shall be a minimum of 5 inches above the bottom of machine.

3.4.11 Piping. Hot and cold water supply piping, valves, and fittings shall have a working pressure of 100 pounds per square inch.

3.4.11.1 Inlet valves. Hot and cold water valves shall be ¾ inch male NPT (National (American) Standard Pipe Taper) thread in accordance with ANSI/ASME B1.20.1 and shall be located on the lower left side of the machine when facing the back of the washer-extractor. If plastic water inlet valves are used, a metal NH (National (American) Standard Hose Coupling) hose coupling with threads in accordance with ANSI/ASME B1.20.7 shall be provided to connect the NPT pipe connector to ship service piping. Inlet connection to the shell shall not be less than 3 inches above the maximum high-water level.

3.4.12 Overflow and vent connections. The washer-extractor shall be fitted with an overflow connection. The overflow connection shall be sized to prevent water from rising 3 inches above the high-water level when both the hot and cold water valves are fully opened and the inlet pressure is 40 pounds per square inch. The shell vent and/or overflow connection shall be configured to prevent any liquid or moisture from discharging from the machine under normal operations. Venting/overflow connection(s) located above the drive belt shall be supported and reinforced to prevent the vent/overflow hose(s) from sagging onto the drive belt in the event moisture accumulates in the vent/overflow hose(s).

3.4.13 Drive. The cylinder shall be V-belt driven by a two-speed electric motor as specified in 3.4.18.1 to provide proper speeds for washing and extracting cycles. The cylinder shall reverse rotation a minimum of three times per minute during the wash cycle.

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3.4.13.1 Bearings. The cylinder shall be supported by replaceable antifriction bearings which shall be provided with replaceable seals to prevent water entry and shall be selected based upon the maximum bearing load during operation of the washer-extractor. The bearings shall be of a readily renewable type and of sufficient capacity and durability to provide vibrationless rotation of the cylinder under full load conditions for not less than 5000 hours.

3.4.13.2 V-belts. V-belts shall have a service life of not less than 5000 hours of machine operation. A means shall be provided for properly tensioning the V-belts.

3.4.13.3 Shafts. Shafts and related drive items shall be constructed so that maximum stresses are not greater than 70 percent of the yield strength of the materials under the maximum applied operating torque during maximum dry weight load capacity operation.

3.4.14 Automatic program control. The washer-extractor shall be provided with an automatic timer-programmer, preprogrammed by the manufacturer with four preset wash formulas: whites, colors, permanent press, and delicates. The automatic program control shall automatically control the water level setting; temperature setting; and the fill, wash, drain, rinse, and extract operations. Unless otherwise specified herein, the control formula shall be programmed to provide not less than one wash, three rinse, two intermediate extract, and one final extract step. The delicate control formula shall be programmed to provide not less than one wash, three rinse, one intermediate extract, and one final extract step.

3.4.14.1 Remote control box. The automatic program control shall be located in a drip proof remote control box. The overall dimensions of the remote control box shall be not greater than 24 inches wide by 9 inches deep by 9 inches high. To suit various shipboard conditions, a minimum of four mounting holes shall be provided in both the bottom of the remote control box for securing the box to a shelf mounted foundation, and in the back of the remote control box for securing the box to a bulkhead mounted foundation. The remote control box shall be provided with not less than a 20 foot length of electrical cable(s) with connectors and receptacles for convenient installation. High and low voltage wires shall be run in separate conduits to avoid electromagnetic interference (EMI). The cable(s) on the control box end shall be hard wired through the bottom of the box with terminal plug(s) on the ends connecting to the washer-extractor. The terminal plug(s) shall have enclosed pin-and-socket connectors and the mating connection shall be a lock type to ensure secure connection. The terminal plug(s) housing shall also be easily unwired/disconnected/readily removable for routing cabling/wiring through ship's stuffing tubes. The terminal plug(s) mating connection shall be located on the lower right side of machine when facing the back of the washer-extractor. Optional knock-outs shall be provided on the sides of the remote control box in order to reroute cabling in the field to suit existing shipboard conditions.

3.4.15 Water level control. An automatic water level control shall be provided with not less than two adjustable water level settings. The water level control shall be electrically interlocked to prevent the washer-extractor program timer from advancing during the water filling operation.

3.4.16 Lubrication. Washer-extractors which require periodic lubrication shall be provided with conveniently located grease fittings that are accessible from the front of the machine.

3.4.17 Finish and color. Unless otherwise specified, the washer-extractor and remote control box finish shall be in accordance with the contractor’s standard commercial practice. Corrosion-resistant steel and galvanized steel surfaces shall not be painted. Welds external to the machine shall be cleaned and ground to present a smooth and uniform finish. Welds internal to the machine shall be cleaned prior to application of any required protective coating.

3.4.18 Electrical requirements. Unless otherwise specified (see 6.2), all electrical equipment shall conform to the requirements in IEEE 45 for electrical shipboard installations. Electrical equipment shall be designed for operation on a Type I electrical system in accordance with MIL-STD-1399-300. Push buttons and switches shall be arranged for safe and convenient operation. The washer-extractor shall operate on 440 volt, 3-phase, 60 Hertz electrical service with a maximum 10-amp fuse. Main electrical feed shall be located on the remote control box.

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3.4.18.1 Motor. Motor shall be two speed and conform to the following requirements:

Voltage:	440 volts, 3-phase, 60 hertz (Hz).
Type:	Squirrel cage induction.
Ambient temperature:	40 °C.
Duty:	Wash: Reversing, intermittent, 3 reversals (minimum) per minute (see 3.4.13).
	Extract: 10 minute full load operation followed by 10 minute rest.
Bearings:	Ball.

3.4.18.2 Controls. Motor controllers shall be of the low voltage protection type. Electrical components mounted outside the control panel shall be provided with drip proof enclosures.

3.4.18.3 Wiring methods and practices. Terminal blocks shall be plainly marked to correspond to the markings on the electrical diagram. Wires run in conduit or harness assemblies shall be marked to correspond with the electrical diagram.

3.4.19 Safety features. Exposed belts, shafts, pulleys, and other moving parts shall be fully enclosed by metal guards. Metal guards shall be firmly supported but readily removable for maintenance.

3.4.19.1 Door interlock. An electric interlock shall be provided to prevent the shell door from opening until the cylinder is at rest following cycle completion and to prevent rotation of cylinder until the shell door is latched closed.

3.4.19.2 Emergency stop. An emergency stop button shall be provided in a readily accessible location on the remote control box for immediately stopping the washer-extractor. The configuration, color, and labeling of the emergency stop button shall be in accordance with the manufacturer’s standard commercial practice.

3.5 Performance. The washer-extractor shall meet the performance requirements in 3.5.1 through 3.5.6.

3.5.1 Inclined operation. When tested as specified in 4.2.3, the washer-extractor shall operate up to and including 30 degrees of inclination without a reduction in performance, spillage shall not exceed a total of one gallon during washer-extractor operation, and spillage shall not cause damage to electrical parts.

3.5.2 Vibration. The washer-extractor shall operate normally and perform its functions both during and after the tests specified in 4.2.4.

3.5.3 Airborne noise and structureborne vibration. The washer-extractor airborne noise shall be not greater than the airborne noise acceptance criteria specified in table II. Unless otherwise specified, the washer-extractor structureborne vibratory acceleration shall be not greater than the acceptance criteria specified on figure 2 of MIL-STD-740-2 for Type III equipment. For washer-extractors that are solidly mounted or Distributed Isolation Material (DIM) mounted shipboard (see 6.2), the 20 dB correction factor specified in the notes on figure 2 of MIL-STD-740-2 shall be subtracted from the Type III curve and the resulting values used as the structureborne vibratory acceleration acceptance criteria.

3.5.4 Extraction efficiency. Water retention following completion of a full wash cycle shall be not greater than 80 percent of the normal dry weight (for instance 16 pounds of test cloths saturated and extracted shall weigh no more than 28.8 pounds) when tested as specified in 4.2.6.

3.5.5 Shock (Type I only). Washer-extractor and remote control box shall meet the Grade B shock requirements of MIL-S-901 when tested as specified in 4.2.8 (see 6.2).

3.5.6 Finished painted surface requirements. Painted surfaces shall be meet the requirements specified in 3.5.6.1 through 3.5.6.3.

3.5.6.1 Hardness requirements. Pencil hardness of painted surfaces shall be greater than 2H when tested as specified in 4.2.9.1.

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3.5.6.2 Adhesion requirements. Painted surfaces shall meet the ASTM D3359 classification 4B acceptance criteria when tested as specified in 4.2.9.2.

3.5.6.3 Humidity requirements. Painted surfaces shall not show any changes in appearance or damage when subjected to the test specified in 4.2.9.3.

3.6 Identification and information plates. Identification and information plates of corrosion resistant steel or brass shall be secured in a conspicuous location on each machine. Marking shall be permanent and legible (see 3.6.1 and 3.6.2).

3.6.1 Marking on identification plate. Marking on the identification plate shall consist of the following:

a. Manufacturer's name, type or model number, serial number, contract or order number, and date of manufacture.

b. Salient design features including size, weight, speeds, voltage, current, and capacity in pounds dry weight per load.

c. PIN and National Stock Number (see 6.2).

3.6.2 Marking on information plate. Information plate shall be marked to include the following: "MACHINE NOT TO BE LOADED WITH MORE THAN RATED POUNDS (DRY WEIGHT) OF CLOTHING."

3.7 Workmanship. Workmanship shall be in accordance with industry practices and shall meet the following:

a. External welds on parts and assemblies provide a smooth and uniform finish. Internal welds were cleaned prior to receiving any required protective coating.

b. All parts are tight and of rigid construction so that the machine does not leak or vibrate during operation.

c. Cylinder both with load and without load does not wobble during operation.

d. Access plates and doors are provided, sized, and located for easy access for maintenance, cleaning, and lubrication.

e. Controls and door handle are suitably located and hazard free to operator.

f. All surfaces, including interior of cylinder, are free of burrs and sharp edges.

g. Safety guards are configured and located to protect operating and maintenance personnel from injury.

h. Drive belt adjustment mechanism is suitable for the purpose intended.

i. Bearings and shafts are properly designed, aligned, and fitted for durable operation.

j. Lubrication fittings are fitted where necessary and accomplish the purpose for which intended.

k. Fastenings accomplish the purpose for which intended.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the washer-extractor when a first article sample is required (see 3.1). First article inspection shall consist of the examinations in table I and tests specified in 4.2.2 through 4.2.10 (see 6.3).

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4.2.1 First article sample. Prior to the start of production and within 30 days after the date of award of the contract or order, one machine of each configuration shall be made ready for approval of the contracting officer or his authorized representative to conduct the tests and examinations specified in table I and tests specified in 4.2.2 through 4.2.10 (see 6.2 and 6.3). Approval of the tested sample by the contracting officer or his authorized representative shall not relieve the supplier of the responsibility of compliance with the requirements of this specification. Type I machines which satisfactorily pass all first article tests do not require retest for Type II applications. Disposition of the first article sample shall be in writing at the direction of the contracting officer or his designated representative (see 6.2 and 6.3).

4.2.1.1 Test conditions. Unless otherwise specified, the test load for the tests specified in 4.2.3 through 4.2.7 shall consist of the washer-extractor maximum dry weight load capacity of test cloths. Test cloths shall be prepared from a white cotton fabric of uniform texture and permanent finish, having a warp of 48, a weft of 48, and weighing not less than 0.33 pound per square yard. Permanent finish, sanforized, white cotton cloth, known as Indianhead cloth may be used. The test cloths shall be cut to 24 by 36 inches and shall be double-hemmed to a finished size of 22 by 34 inches. The dry weight of test cloths shall be determined at a temperature of 70±2 °F. The ambient conditions where the tests are conducted shall be 75±10 °F and 60±20 percent relative humidity.

4.2.2 Disassembly. Prior to conducting the inspection tests, the sample machine shall be disassembled to the extent required and the component parts passed through a 30-inch diameter by 60-inch long cylinder. Machine shall then be reassembled for testing.

TABLE I. Classification of defects.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Materials	Materials not as specified (see 3.3).	X	
Configuration, general	Configuration not as specified (see 3.4 and 3.7).	X	
Special Design Requirements	Not as specified (see 3.4.1)	X	
Automatic program control	Does not provide control as specified (see 3.4.14).	X	
Water level controls	Not as specified (see 3.4.15).	X	
Door interlock	Not provided, not as specified (see 3.4.19.1).	X	
Emergency stop	Not provided, not as specified (see 3.4.19.2).	X	
Capacity, size, weight, water usage	Machine dry weight load capacity not as specified (see 3.4.1.1).	X	
	Machine exceeds specified dimensions (see 3.4.2 and 3.4.14.1).	X	
	Machine exceeds weight limits (see 3.4.3).	X	
	Machine water consumption exceeds limits (see 3.4.4).	X	

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TABLE I. Classification of defects - Continued.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Construction/mechanical	Mounting holes not provided, not as specified (see 3.4.5).	X	
	Mounting bolts not provided, not as specified (see 3.4.5)		X
	Shell not as specified (see 3.4.6)	X	
	Shell openings not as specified (see 3.4.6.1).	X	
	Cylinder not as specified (see 3.4.7 and 3.7.c.)	X	
	Cabinet not as specified (see 3.4.8 and 3.7)	X	
	Door not as specified (see 3.4.9)	X	
	Drain valve not as specified (see 3.4.10).	X	
	Piping, valves, and fittings not as specified (see 3.4.11 and 3.4.11.1).	X	
	Inlet valves are not 3 inches minimum above the maximum high water level (see 3.4.11.1).	X	
	Overflow/vent connection(s) not as specified (see 3.4.12).	X	
	Drive not as specified (see 3.4.13 and 3.7)	X	
	Cylinder bearings and seals not as specified (see 3.4.13.1 and 3.7).	X	
	V-belts not properly sized or not readily adjustable (see 3.4.13.2 and 3.7).	X	
	Shaft not as specified (see 3.4.13.3 and 3.7)	X	
	Remote control box not as specified (see 3.4.14.1)	X	
	Grease fittings not as specified (see 3.4.16 and 3.7).	X	
	Finish and color not as specified (see 3.4.17)	X	
	Metal guards not as specified (see 3.4.19 and 3.7)	X	

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TABLE I. Classification of defects - Continued.

Examine	Defect (requirement paragraph)	Classification	
		Major	Minor
Construction/electrical	Electrical requirements not as specified (see 3.4.18)	X	
	Motor not as specified (see 3.4.18.1).	X	
	Controls not as specified (see 3.4.18.2)	X	
	Electrical connection boxes are not drip proof enclosed or better (see 3.4.18.2).	X	
	Electrical terminal blocks not marked to correspond with electrical diagram (see 3.4.18.3).	X	
	Wires not properly marked to correspond with electrical diagram (see 3.4.18.3).	X	
	Length of remote control box cable not as specified (see 3.4.14.1).	X	
Safety features	Belts, shafts, pulleys not properly enclosed (see 3.4.19).	X	
Identification and information plates	Material not as specified (see 3.6).		X
	Marking not permanent or legible (see 3.6).	X	
	Marking on identification and information plates not provided, not as specified (see 3.6.1 and 3.6.2).	X	
Workmanship	External welds not ground and cleaned (see 3.7).		X
	Surfaces of cylinder are not free of burrs and sharp edges (see 3.4.7).	X	
	Surfaces other than cylinder are not free from burrs and sharp edges (see 3.7).		X
	Fasteners/nuts not suitable for intended purpose (see 3.3.5 and 3.7)	X	

4.2.3 Inclination test. The machine shall be inclined and operated at an angle of 30 degrees to each side of two vertical planes at right angles to each other. The machine shall be loaded as specified in 4.2.1.1 and operated for a full cycle in each of the four positions. During each of the four tests, a means shall be provided to catch any spillage that occurs during washer-extractor operation. Following each of the four tests, the spillage shall be measured, and the spillage amount and whether any electrical parts were damaged by the spillage shall be recorded.

4.2.4 Vibration tests. The machine shall be tested without packaging or crating.

4.2.4.1 Vibration testing. The machine shall be loaded as specified in 4.2.1.1 and subjected to a Type I vibration test in accordance with MIL-STD-167-1, except the variable frequency tests are not required.

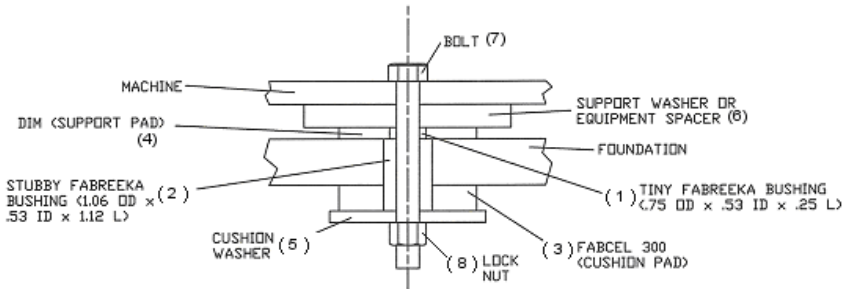
4.2.5 Airborne noise and structureborne vibration tests. Unless otherwise specified, the machine airborne noise and structureborne vibration acceleration tests shall be conducted in accordance with MIL STD-740-1 and MIL-STD-740-2, respectively. The machine shall be tested on the NSWCCD Code 9840 mounting system in lieu of the low frequency resilient mounts specified in 5.5.1 of MIL-STD-740-1 and 5.6.1 of MIL-STD-740-2. A sketch of the NSWCCD mounting system and a parts list are provided on figure 1.

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4.2.5.1 Test fixture and test set up. The washer-extractor shall be mounted on a test fixture to simulate the most severe mounting condition likely encountered in an actual shipboard installation. The test fixture shall be approved by the accepting government activity prior to conducting the tests. The mounting boxes used to mount the washer-extractor to the test fixture shall be as specified in 4.2.5.5. The machine shall be installed on the test fixture as specified in 4.2.5.6.

4.2.5.2 Test conditions. Two sets of airborne noise and structureborne vibration tests shall be conducted. The first set of tests shall be conducted with the machine loaded with 10 pounds (dry weight) of test cloths. The second set of tests shall be conducted with the machine loaded as specified in 4.2.1.1. Airborne noise and structureborne vibration shall be monitored in all modes of machine operation. The noise level plots shall include the wash, rinse, extract, and fill modes with both inlet valves open.

4.2.5.3 Airborne noise test. The washer-extractor shall be mounted as specified in 4.2.5.1 and airborne noise tested in each of the test conditions specified in 4.2.5.2 using the procedures for sound pressure level measurements.



Item no.	Description	Material	Thickness	Length	Outside diameter	Inside diameter
1	Stubby Fabreeka bushing	Fabreeka	0.110	0.250	0.750	0.531
2	Tiny Fabreeka bushing	Fabreeka	0.265	1.125	1.062	0.531
3	Cushion pad	Fabcel 300	0.500	NA	2.312	1.062
4	Support pad	DIM	0.328	NA	2.312	0.750
5	Cushion washer	Steel	0.250	NA	2.650	0.531
6	Support washer	Steel	0.625	NA	2.650	0.531
7	Bolt - ½”-13UNC	Steel	NA	3.750	0.500	NA
8	Lock nut – ½”	Steel	NA	NA	NA	0.500

FIGURE 1. NSWCCD mounting system and parts list.

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4.2.5.3.1 Sound pressure acceptance levels. The sound pressure level measurements shall be not greater than the octave band sound pressure levels and overall dBA level specified in table II.

TABLE II. Washer-extractor airborne noise acceptance levels (maximum).

Sound pressure, dB at 1 meter RE .0002 dyne/cm^2 for octave band center frequencies, Hz									Sound pressure, dBA at 1 meter RE 0.0002 dyne/cm^2
32	63	125	250	500	1K	2K	4K	8K	
73	69	71	63	62	62	59	58	59	66

4.2.5.4 Structureborne vibration test. The washer-extractor shall be mounted as specified in 4.2.5.1 and structureborne vibration tested in each of the test conditions specified in 4.2.5.2.

4.2.5.4.1 Selection of transducer locations. Transducer triaxial blocks shall be attached to the supporting members of the machine at not less than two diagonally opposite corners as close as possible to the machine mounting bolts. Not less than one of the corners selected shall be the one nearest the attachment for the drive motor.

4.2.5.4.2 One-third octave band measurement. The noise level reported for each band shall be the maximum level appearing (excluding spurious signals) in any direction for each of the machine's acoustically significant operational modes (including wash, rinse, extract, and fill modes).

4.2.5.5 Test fixture mounting boxes. The test fixture equipment mounting boxes shall be fabricated and installed on the test fixture as specified on figure 2 and figure 3.

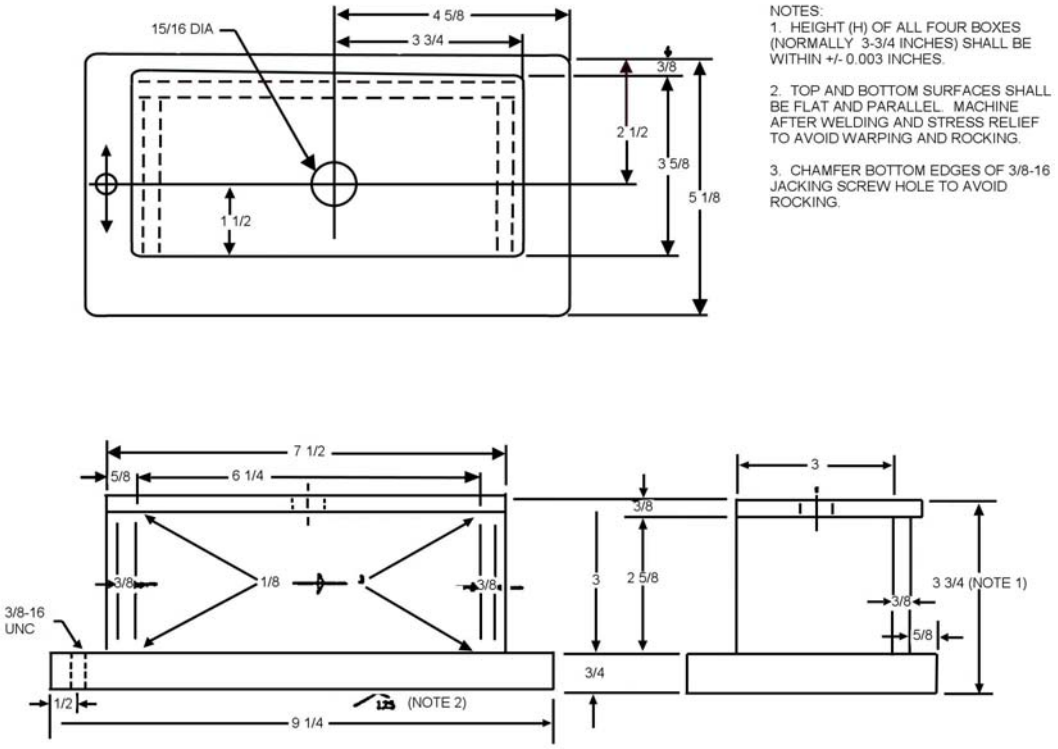


FIGURE 2. Mounting boxes – steel, 4 – required.

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4.2.5.6 Test set up. The washer-extractor shall be mounted on the test fixture in accordance with the following requirements:

a. The washer-extractor shall be attached to the mounting boxes, figure 2, using the NSWCCD mounting system, figure 1.

b. Machine feet should be level and in place within 0.015 inch before inserting rubber pads.

c. Hold down nuts shall be tightened evenly and gradually to compress the resilient pads to a final installed height that is 73 to 75 percent of the pad initial free height.

d. The mounting boxes shall be firmly attached to a structural floor or machinery bedplate.

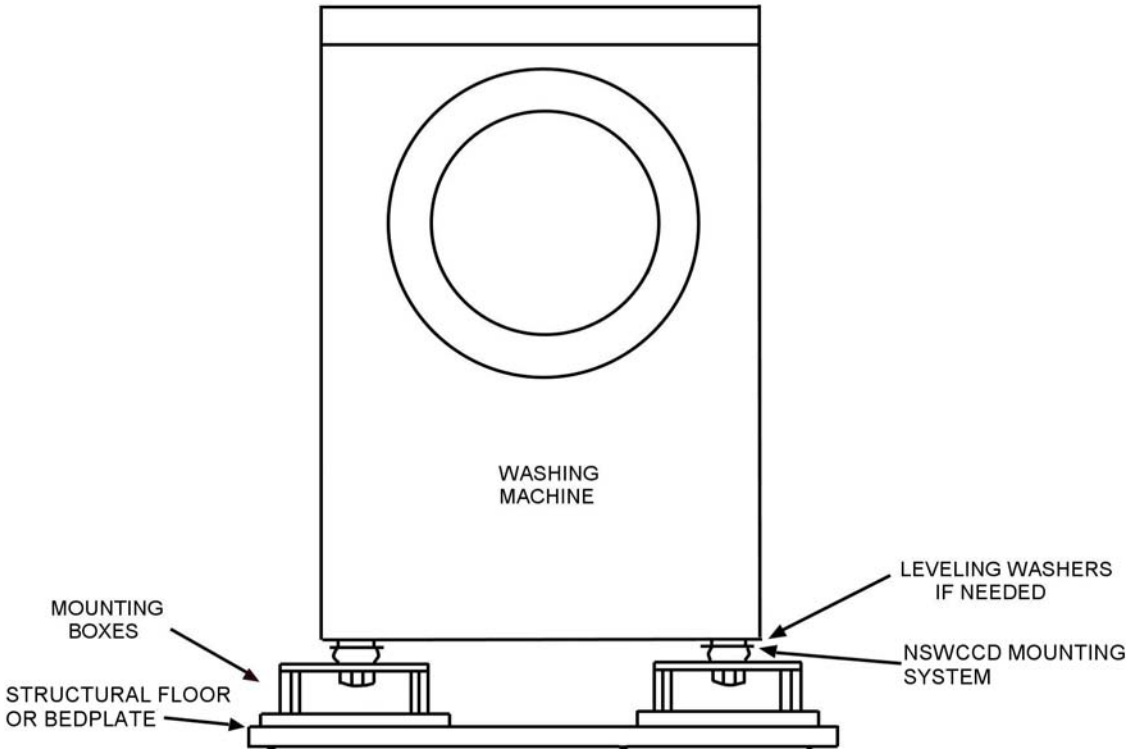


FIGURE 3. Test fixture and item assembly.

4.2.6 Extraction efficiency test. The machine shall be operated with the test load specified in 4.2.1.1. After each full operation, the test load shall be weighed within 3 minutes of the cycle completion and the weight recorded. An average of 10 runs shall be used to determine the extraction efficiency of the machine.

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4.2.7 Door interlock, emergency stop, and water level control test. The door interlock shall be tested with the machine operating in the fill, wash, rinse, and extraction modes; with the machine stopped during the fill, wash, rinse, and extraction modes; and at machine cycle completion. For door interlock tests conducted with the machine stopped during the fill, wash, rinse, and extraction modes; the machine shall be stopped using the emergency stop and the door interlock tested once the cylinder is at rest. Following each test, the emergency stop shall be reset and the machine restarted. The emergency stop shall immediately stop the machine when activated. The door shall open only when the cylinder is at rest following cycle completion. Once the door interlock and emergency stop tests are complete, the machine shall be set to the maximum water level and operated with the test load specified in 4.2.1.1. After the wash cycle starts, the machine shall be stopped and the water allowed to settle. Both the high water level and a vertical distance 3 inches above the high water level shall be marked for reference during the tests. The machine shall be restarted and complete the full test cycle. After completion of the first test cycle, the machine loaded as specified in 4.2.1.1 shall be set at the maximum water level with both the hot and cold inlet valves fully open. With a water inlet pressure of 40 pounds per square inch, the machine shall be started. The start time and length of each mode of machine operation shall be recorded for comparison to the manufacturer's programmed control formula to ensure each wash, rinse, and extraction mode adhered to the control formula length. In addition, the water level shall be visually observed throughout the test to ensure that the maximum water level was obtained before the wash and rinse modes began operation, and that the water level never exceeded the 3-inch mark above the high water level during machine operation.

4.2.8 High impact shock test (Type I only). One washer-extractor and one remote control box shall be subjected to a shock test in accordance with MIL-S-901 as follows:

a.	Shock grade	-	Grade B
b.	Equipment class	-	Class I
c.	Test category	-	The washer-extractor shall be subjected to a mediumweight shock test. The remote control box shall be subjected to a lightweight shock test.
d.	Shock test type	-	Type A
e.	Failure definition	-	Any part of washer-extractor or remote control box coming adrift or separating from the test items constitutes failure of test.
f.	Mounting	-	Washer-extractor shall be tested on the fixture shown on figure 16 of MIL-S-901. The remote control box shall be tested on fixture 4-A as shown on figure 7 of MIL-S-901.
g.	Mode	-	Equipment shall not be energized or operating during the test.
h.	Disposition	-	Equipment to be disposed of by the contractor, and shall not be delivered as part of the contract.
i.	Quantity for test	-	1 each of washer-extractor and remote control box.

4.2.9 Test panel description. Four test panels shall be fabricated of cold rolled steel sheet. Test panels shall be 0.0239 inch (0.61 mm) thick, not less than 4½-inches (114 mm) wide, and not greater than 7½-inches (191 mm) long. The four test panels shall be pretreated, prime coated, and painted in the same manner as that used for painting the driving trunnion, spider, remote control box, and structure. Test panels shall be tested as specified in 4.2.9.1 through 4.2.9.3.

4.2.9.1 Hardness. One test panel shall be subjected to a pencil hardness test in accordance with ASTM D3363. Forward pencil strokes shall be not less than ¼-inch (6.4 mm) long.

4.2.9.2 Adhesion. Two test panels shall be subjected to a cross hatch test in accordance with ASTM D3359 Test Method B to determine the adhesion between the paint and primer and between the primer and base metal. The test panels shall not be subjected to preliminary testing or immersion prior to determining the adhesion rating.

4.2.9.3 Humidity. One test panel shall be exposed to 100 percent relative humidity at 120 °F (49 °C) for 168 hours. After 168 hours exposure, the test panel shall immediately be subjected to the pencil hardness test specified in 4.2.9.1.

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4.2.10 Insulation test. The insulation resistance of motor, automatic program control, and connecting cable shall be measured by a 500-volt direct current insulation resistance tester. Insulation resistance for each component shall be not less than 10 megohms.

4.3 Conformance inspection. Conformance inspection shall consist of examinations of table I and the tests specified in 4.3.4. Records of the inspections, tests, and any resulting rejections shall be maintained by the contractor for a period of 3 years following contract completion.

4.3.1 Lot. For the purpose of conformance inspection and test sampling, a lot is defined as all washer-extractor machines of the same design, produced in the same facility, using the same production processes and materials, and being offered for delivery at the same time.

4.3.2 Sampling for examination. A random sample of machines shall be selected from each lot of completed units in accordance with table III and visually examined for defects in accordance with table I. If one or more major defects are found in any sample, the entire lot shall be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be examined in accordance with the sampling plan contained herein.

Table III. Sampling for examination and operational tests.

Lot size	Sample size	
	Examinations	Operational testing
2 to 8	All	3
9 to 25	8	3
26 to 50	8	5
51 to 90	8	6
91 to 150	12	7
151 to 280	19	10
281 to 500	21	11
501 to 1200	27	15
1201 to 3200	35	18

4.3.3 Sampling for tests. A random sample of machines shall be selected in accordance with table III from each lot which passes the examinations of 4.3.2. The random sample shall be tested in accordance with 4.3.4. If any machine fails any of the tests specified in 4.3.4, the entire lot shall be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be examined and tested in accordance with the sampling plan contained herein.

4.3.4 Lot tests. Each sample machine selected in accordance with 4.3.3 shall be subjected to the tests specified in 4.3.4.1 through 4.3.4.3.

4.3.4.1 Operational testing. Each machine shall be connected to an electrical power supply and water supply, and operationally tested as follows:

a. Automatic program control automatically controls all sequences of operation in proper sequence for three full cycles.

b. Door electric interlock prevents the door from opening until the cylinder is at rest following cycle completion and prevents rotation of cylinder until the door is latched closed.

c. Piping, shell, outflow/vent connection(s), and door components exhibit no leakage during or following machine operation.

d. Cylinder reverses not less than three times per minute.

e. Cylinder does not wobble during operation.

f. Machine does not leak or vibrate during operation.

g. Emergency stop immediately stops the machine when activated.

h. Indicating lights, switches, push buttons, and timers function properly.

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4.3.4.2 Insulation test. The insulation resistance of motor, automatic program control, and connecting cable shall be measured by a 500-volt direct current insulation resistance tester. Insulation resistance for each component shall be not less than 10 megohms.

4.3.4.3 Power input. Motors shall not draw in excess of rated current after accelerating to full speed.

4.4 Inspection of packaging. Sample packaging and packs and the inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents therein.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Commercial laundry washer-extractor machines acquired under this specification are intended for submarine service. Type I washer-extractors are for use in applications requiring underwater explosion shock resistance and may also be used in Type II washer-extractor applications. Type II washer-extractors are for use only in applications that do not require underwater explosion shock resistance. Commercial machines offered to the Government may require alteration to meet the specification requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Type required (see 1.2)
c. First article, when required (see 3.1).
d. Capacity of machine required (see 3.4.1.1)
e. Electrical requirements if other than as specified (see 3.4.18).
f. Whether washer-extractor is solidly mounted or DIM mounted shipboard (see 3.5.3).
g. Shock test, when required (see 3.5.5).
h. Part Identification Number and National Stock Number (see 3.6.1.c).
i. First article preparation schedule (see 4.2.1).
j. Disposition of first article sample (see 4.2.1).
k. Packaging (see 5.1).
l. Provisioning (see 6.4).
m. Quantity required.

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, or a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested (see 4.2.1). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Subject term (key word) listing.

Automatic control

Interlock

Reversing

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 3510-2008-005)

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